

Encryption in transit for Google Cloud



This content was last updated in April 2025, and represents the status quo as of the time it was written. Google's security policies and systems may change going forward, as we continually improve protection for our customers.

At Google, our security controls help protect your data—whether it is traveling over the internet, moving within Google's infrastructure, or stored on our servers. Central to Google's security strategy are authentication, integrity, and encryption, for both data at rest and data in transit. This paper describes how we designed Google Cloud to encrypt data in transit from the internet and data in transit within Google's networks. This document doesn't apply to data in transit over interconnects between customer data center networks and Google's data center networks.

Encryption in transit uses various technologies to help protect data, including transport layer security (TLS), BoringSSL, application layer transport security (ALTS), and the [PSP security protocol](#)

(<https://cloud.google.com/blog/products/identity-security/announcing-psp-security-protocol-is-now-open-source>)

. In addition to the default protection that Google provides, you can further protect data by adding encryption options such as IPsec, managed TLS certificates, and Cloud Service Mesh.

This document is aimed at CISOs and security operations teams using or considering Google Cloud. This document assumes a basic understanding of [encryption](#)

(<https://en.wikipedia.org/wiki/Encryption>) and [cryptographic primitives](#)

(https://en.wikipedia.org/wiki/Cryptographic_primitive).

Authentication, integrity, and encryption

Google employs the following security measures to help ensure the authenticity, integrity, and privacy of data in transit:

- **Authentication** verifies the identity of a peer (either a human or a process) in a connection.

- **Integrity** prevents data that you send from being altered while in transit between the source and the destination.
- **Encryption** uses cryptography to make your data unreadable while in transit and keep it confidential.

Encryption in transit helps protect your data if communications are intercepted while data moves between the end user and Google Cloud or between two services. Encryption in transit authenticates the endpoints and encrypts the data before transmission. On arrival, the receiver decrypts the data and verifies that it was not modified during transit.

Encryption is one component of a broader security strategy. Encryption in transit defends your data against potential attackers and removes the need for Google, Google Cloud customers, or end users to trust the lower layers of the network.

How traffic gets routed

This section describes how requests get from an end user to the appropriate Google Cloud service or customer application, and how traffic is routed between services.

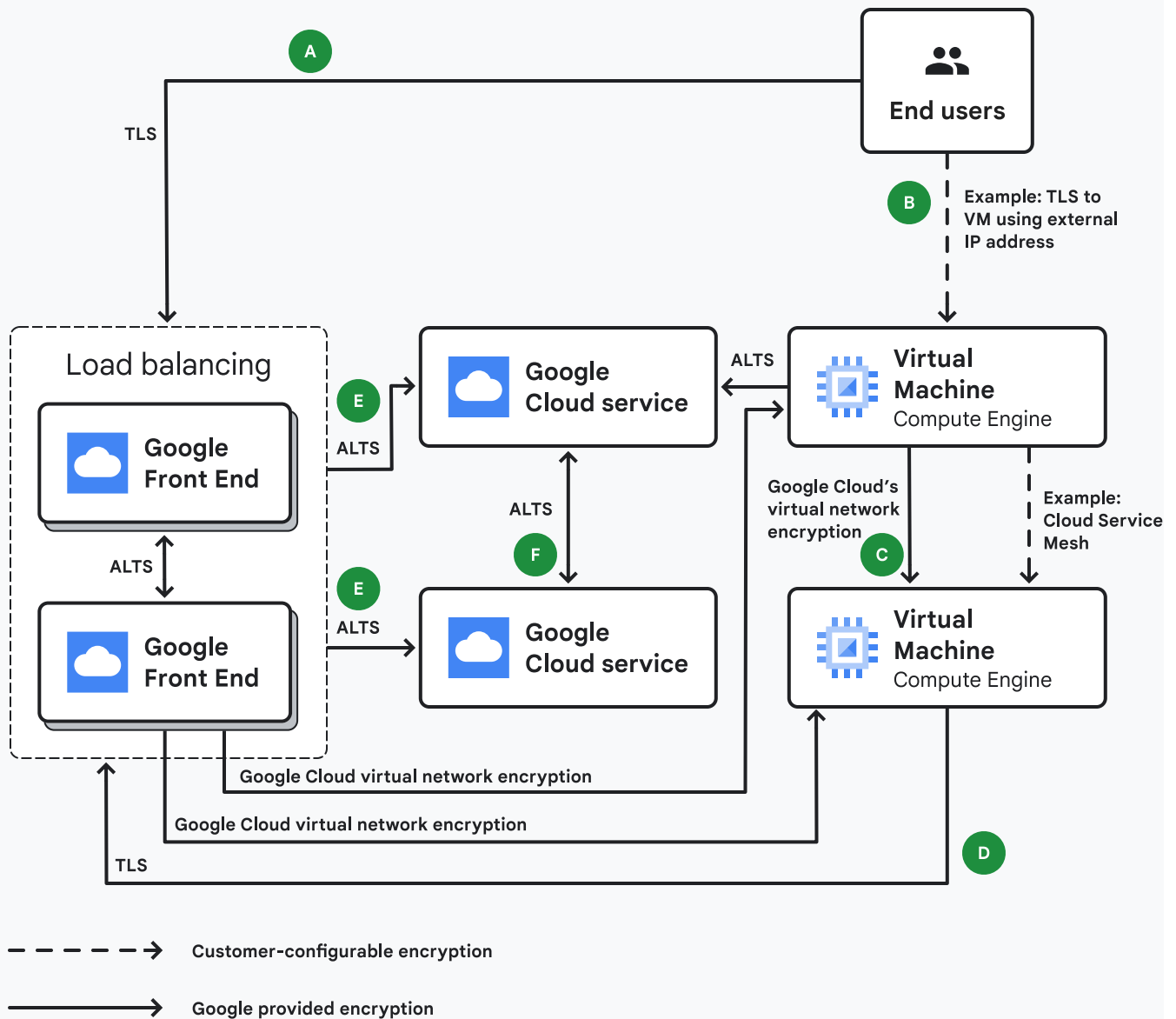
A *Google Cloud service* is a modular cloud service that we offer to our customers. These services include compute, data storage, data analytics, and machine learning. For example, Cloud Storage is a Google Cloud service.

A *customer application* is an application hosted on Google Cloud that you, as a Google customer, can build and deploy using Google Cloud services. Customer applications or partner solutions that are hosted on Google Cloud are not considered Google Cloud services. For example, an application you build using Cloud Run, Google Kubernetes Engine, or a VM in Compute Engine is a customer application.

The following diagram shows traffic paths from the end user to Google, paths within Google's network, and the security for each connection. The following traffic paths are shown:

- End user on the internet to a Google Cloud service (labeled A in the diagram)
- End user on the internet to a customer application that is hosted on Google Cloud (labeled B in the diagram)
- Virtual machine to virtual machine (labeled C in the diagram)
- Virtual machine to Google Front End (GFE) (labeled D in the diagram)

- GFE to Google APIs and services (labeled E in the diagram)
- Google Cloud service to Google Cloud service (labeled F in the diagram)



Encryption in transit between the end user and Google

The following sections provide more detail about the end-user routing requests that are shown in the preceding diagram.

End user to a Google Cloud service

Google Cloud services such as Cloud Storage or Compute Engine are cloud services that we offer to customers and run in our production environment

(</docs/security/production-services-protection>). Google Cloud services accept requests from around the world using a globally distributed system called Google Front End (GFE)

(</docs/security/infrastructure/design#google-frontend-service>). GFE terminates traffic for incoming HTTP(S), TCP, and TLS connections; provides DDoS attack countermeasures; and routes and load balances traffic to the Google Cloud services themselves. GFE points of presence exist around the globe with paths that are advertised using unicast or Anycast

(<https://tools.ietf.org/html/rfc1546>).

GFE routes traffic from an end user over Google's network to a Google Cloud service, and from an end user to a customer application that is hosted on Google Cloud and uses Cloud Load Balancing (</load-balancing/docs/load-balancing-overview>).

Requests that clients send to a Google Cloud service over HTTPS, HTTP/2, or HTTP/3 are secured with TLS. TLS in the GFE is implemented with BoringSSL

(<https://boringssl.googlesource.com/boringssl>), a Google-maintained, open-source implementation of the TLS protocol. BoringCrypto, the core of BoringSSL, is validated to FIPS 140-3 level 1 (<https://csrc.nist.gov/projects/cryptographic-module-validation-program/certificate/4735>).

The GFE negotiates industry-standard cryptographic parameters with the client, including forward-secure key negotiation. For older, less capable clients, the GFE chooses the strongest legacy cryptographic primitives that the client offers.

End user to a customer application hosted on Google Cloud

You can route end user traffic from the internet to your customer applications that are hosted on Google Cloud using a direct connection or through a load balancer. When the traffic is routed directly, the traffic is routed to the external IP address of the VM server that hosts the application.

You can use TLS with the external Application Load Balancer or the external proxy Network Load Balancer to connect to your application running on Google Cloud. When you use a Layer 7 load balancer, the connection between the end user and the load balancer is encrypted using TLS by default (provided that the end user's client application supports TLS). GFE terminates the TLS connections from your end users using TLS certificates that are self-managed or Google-managed. For more information about customizing your certificate, see SSL certificates overview (</compute/docs/load-balancing/http/ssl-certificates>). To enable encryption between the load balancer and the VM that hosts the customer application, see Encryption

from the load balancer to the backends

(/load-balancing/docs/ssl-certificates/encryption-to-the-backends).

To configure mutual TLS (mTLS), see Mutual TLS overview (/load-balancing/docs/mtls). For workloads on GKE and Compute Engine, consider Cloud Service Mesh (/service-mesh/docs/security/security-overview) so that you can use mTLS for mutual authentication (client and server) and encrypt communications in transit using certificates that you manage.

For Firebase Hosting (https://firebase.google.com/docs/hosting/custom-domain) and Cloud Run custom domains (/run/docs/mapping-custom-domains), consider our free and automated TLS certificates. With Cloud Run custom domains, you can also provide your own SSL certificates (/load-balancing/docs/https/setup-global-ext-https-serverless) and use an HTTP strict transport security (HSTS) header.

Encryption in transit within Google networks

Google Cloud encrypts customer data in transit within Google's networks, unless described otherwise in this section.

Specialized ultra-high performance interconnects that connect TPUs or GPUs within Google's AI supercomputers are separate from the networks described in this document. In Google Cloud, these ultra-high performance interconnects are ICI for TPUs and NVLink for GPUs. For more information, see TPU architecture (/tpu/docs/system-architecture-tpu-vm) and GPU machine types (/compute/docs/gpus).

Traffic over connections to VMs using external IP addresses leaves Google's networks. You are responsible for configuring your own encryption for those connections.

Google Cloud virtual network authentication and encryption

Google Cloud's virtual network encrypts, integrity-protects, and authenticates traffic between VMs.

Encryption of private IP traffic within the same VPC or across peered VPC networks within Google Cloud's virtual network is performed at the network layer.

Each pair of communicating hosts establishes a session key using a control channel that is protected by ALTS (#service-to-service-authentication) for authenticated, integrity-protected, and encrypted communications. The session key is used to encrypt VM-to-VM communication between those hosts, and session keys are rotated periodically.

VM-to-VM connections within VPC networks and peered VPC networks inside of Google's production network are integrity-protected and encrypted. These connections include connections between customer VMs and between customer and Google-managed VMs such as Cloud SQL. The diagram in How traffic gets routed (#how-traffic) shows this interaction (labeled connection C). Note that because Google activates automatic encryption based on the use of internal IP addresses, connections between VMs using external IP addresses aren't automatically encrypted.

Google Cloud's virtual network authenticates all traffic between VMs. This authentication, achieved using security tokens, prevents a compromised host from spoofing packets on the network.

Security tokens are encapsulated in a tunnel header which contains authentication information about the sender and receiver. The control plane on the sending host sets the token, and the receiving host validates the token. Security tokens are pre-generated for every flow, and consist of a token (containing the sender's information) and the host secret.

Connectivity to Google APIs and services

Traffic handling differs depending on where the Google Cloud service is hosted.

Most Google APIs and services are hosted on GFEs. VM to GFE traffic uses external IP addresses to reach Google Cloud services, but you can configure private access (/vpc/docs/private-access-options) to avoid using external IP addresses. The connection from the GFE to the service is authenticated and encrypted.

Some services, such as Cloud SQL (/sql/docs/mysql/introduction), are hosted on Google-managed VM instances. If customer VMs access the services hosted on Google-managed VM instances using external IP addresses, traffic remains in Google's production network but isn't automatically encrypted because of the use of the external IP addresses. For more information, see Google Cloud virtual network authentication and encryption (#gcp-virtual).

When a user sends a request to a Google Cloud service, we secure the data in transit (providing authentication, integrity, and encryption) using HTTPS with a certificate from a web

(public) certificate authority. Any data the user sends to the GFE is encrypted in transit with TLS or QUIC. GFE negotiates a particular encryption protocol with the client depending on what the client can support. GFE negotiates more modern encryption protocols when possible.

Service-to-service authentication, integrity, and encryption

Google's infrastructure uses ALTS for the authentication, integrity, and encryption of connections from the GFE to a Google Cloud service, and from one Google Cloud service to another Google Cloud service.

ALTS uses service-based identities

([/docs/security/binary-authorization-for-borg#step_4_service-based_identity](https://docs/security/binary-authorization-for-borg#step_4_service-based_identity)) for authentication. Services running in Google's production environment are issued credentials asserting their service-based identities. When making or receiving RPCs from other services, a service uses its credentials to authenticate. ALTS verifies that these credentials are issued by Google's internal CA. Google's internal CA is unrelated and independent of the Certificate Authority Service (<https://cloud.google.com/security/products/certificate-authority-service>).

ALTS uses encryption and cryptographic integrity protection for traffic that carries Google Cloud data from the GFE to a service and between services that are running in Google's production environment.

ALTS is also used to encapsulate other Layer 7 protocols, such as HTTP, in RPC mechanisms for traffic between GFEs and Google Cloud services. This protection helps isolate the application layer and removes any dependency on the network path's security.

Encryption in transit methods

The following sections describe some of the technologies that Google uses to encrypt data in transit.

Network encryption using PSP

PSP is a transport-independent protocol that enables per-connection security and supports offloading of encryption to smart network-interface card (SmartNIC) hardware. Whenever SmartNICs are available, ALTS uses PSP to encrypt data in transit across our network.

PSP is designed to meet the requirements of large-scale data-center traffic. Google infrastructure uses PSP to encrypt traffic within and between our data centers. PSP also supports non-TCP protocols such as UDP and uses a unique encryption key for each connection.

Application layer transport security

ALTS is a mutual authentication and encryption system developed by Google. ALTS provides authentication, confidentiality, and integrity for data-in-transit between services running in Google's production environment. ALTS consists of the following protocols:

- **Handshake protocol:** The client initiates a combined elliptic curve and quantum-safe key exchange. At the end of the handshake, involved services authenticate each other's identities by exchanging and verifying signed certificates and compute a shared traffic key. Among the algorithms that are supported for deriving the shared traffic key is the NIST post-quantum algorithm [ML-KEM \(FIPS 203\)](https://csrc.nist.gov/pubs/fips/203/final) (<https://csrc.nist.gov/pubs/fips/203/final>). For more information, see [Post-quantum cryptography](https://cloud.google.com/security/resources/post-quantum-cryptography) (<https://cloud.google.com/security/resources/post-quantum-cryptography>).
- **Record protocol:** Service-to-service data is encrypted in transit using the shared traffic key. By default, ALTS uses AES-128-GCM encryption for all traffic. Data in transit within Google's lowest-level storage system uses AES-256 encryption, and ALTS provides AES message authentication. Encryption in ALTS is provided by BoringSSL or PSP. This encryption is validated at FIPS 140-2 level 1 or FIPS 140-3 level 1.

The root signing keys for ALTS certificates are stored in Google's internal CA.

What's next

- For information about Google's post-quantum cryptography resources, see [Post-quantum cryptography](https://cloud.google.com/security/resources/post-quantum-cryptography) (<https://cloud.google.com/security/resources/post-quantum-cryptography>).
- For more information about our data center security, see [data center security](https://www.google.com/about/datacenters/data-security/) (<https://www.google.com/about/datacenters/data-security/>).
- For information about security configuration options for interconnects between Google Cloud and customer data center networks, see [Choosing a Network Connectivity product](#)

([IPSec](#)) (</network-connectivity/docs/how-to/choose-product>) and [MACsec for Cloud Interconnect overview](#) (</network-connectivity/docs/interconnect/concepts/macsec-overview>).

- For information about Google Cloud compliance and compliance certifications, see the [compliance resource center](#) (<https://cloud.google.com/compliance>), which includes our [SOC 3 audit report](#) (<https://www.google.com/work/soc3.html>).
- For best practices on how to secure your data in transit, see the [Enterprise foundations blueprint](#) (</architecture/security-foundations/networking>), [Google Cloud Architecture Framework: Security, privacy, and compliance](#) (</architecture/framework/security>), and [Decide how to meet regulatory requirements for encryption in transit](#) (</architecture/landing-zones/decide-security#encrypt-transit>).

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